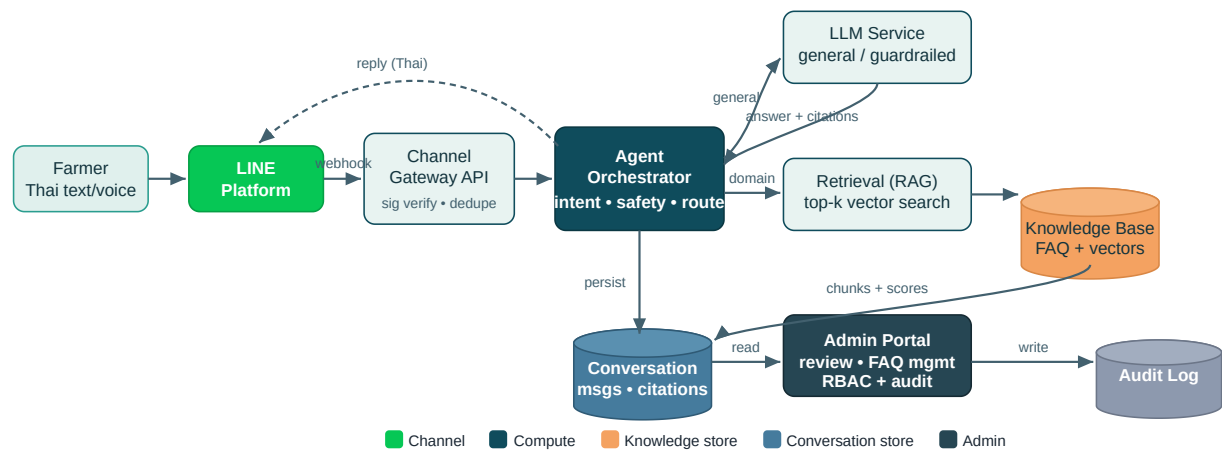


Thai Farmer LINE LLM Advisory Agent — Data Architecture

Farmers chat in Thai over LINE with a guardrailed LLM agent. Domain questions are answered **only** from a curated FAQ knowledge base (RAG) with citations; admins review every interaction.

Version 0.1 (Draft) • Status: For review • 2026-07-16 • Owner: Platform Engineering

System & Data Flow



Anti-Misinformation Controls (data-level)

- **Grounded-only answers:** domain replies must carry ≥ 1 citation; else blocked.
- **Confidence gate:** retrieval score below threshold $\rightarrow$ clarify/refuse, never hallucinate.
- **High-risk blacklist:** pesticide dosage, medical, legal, finance $\rightarrow$ safe template + escalate.
- **Published-only retrieval:** only `status=published` within `valid_from/to`.
- **Source attribution:** authoritative `source` (e.g. DOA/RD) surfaced in answer.

Data Stores

Store	Tech	Holds
Relational	PostgreSQL	identity, consent, sessions, messages, FAQ meta
Vector	pgvector / Qdrant	FAQ chunk embeddings
Cache	Redis	session context, dedupe, rate-limit
Object	S3 / MinIO	media, exports
Audit	append-only + WORM	admin access trail

Core Data Entities

Entity	Key fields
farmer	id, line_user_id (pseudonymous), province, preferred_lang, status
consent	farmer_id, purpose(service/analytics/contact), granted, version
session	farmer_id, channel, context_summary, state(JSONB)
message	session_id, direction, role, content, intent, route, model, tokens, confidence, latency

Entity	Key fields
faq_doc	title, category, body_th, source, status, version, valid_from/to
faq_chunk	faq_doc_id, text, embedding(vector), hash
citation	message_id → faq_chunk_id, score
audit_event	actor, action, target, metadata (append-only)

Message Routes

- smalltalk general chat via guardrailed LLM.
- faq_grounded answered from KB with citations.
- refused weak match / high-risk → safe response.
- escalated handed to admin / human.

Retention & Lifecycle

Data	Hot	Then
Message content	90d	archive 1y → delete
Media	30d	delete w/ message
Session (Redis)	30m idle	auto-expire
FAQ docs	versioned	archive on deprecate
Audit	1–2y	WORM

Governance — PDPA (Thailand)

- Consent:** purpose-based, versioned; captured at LINE onboarding.
 - Minimization:** store pseudonymous line_user_id; no phone/national ID.
 - Erasure:** cascade delete messages/media/embeddings; audit retained lawfully.
 - Access/RBAC:** roles viewer/editor/dpo; PII view is logged.
 - Residency:** prefer in-region LLM+stores; redact PII before external model calls.
- Key risks → hallucinated agronomic advice (→ grounded-only + citations); stale seasonal FAQ (→ validity dates + review); PII leak to external LLM (→ redaction + DPA); webhook replay (→ signature + idempotency).

Non-Functional Highlights

Idempotency	Latency	Observability	Availability	Backup/DR
Dedupe on line_message_id; retries never double-store	Retrieval+gen p95 within LINE reply window; cache hot FAQs	tokens, latency, route, confidence per message	Stateless gateway/orchestrator; DB replicas; Redis context	Postgres PITR; versioned object store; restore tests

Open questions: LLM hosting & data residency • voice/STT scope • human-in-the-loop escalation model • multi-tenant vs single-tenant • legal audit retention period.